Enhancing Clinical Outcome Predictions through Effective Sample Size Evaluation in Graph-Based Digital Twin Modeling

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Introduction

Digital twins in healthcare revolutionize personalized diagnosis, prognosis, and treatment. SynTwin, a novel methodology that combines synthetic data and network science, has shown promise in predicting breast cancer mortality, advancing precision medicine. In this study, we further validate SynTwin by applying it to various cancer types from the SEER program, assessing its efficacy across sample sizes (1,000 to 30,000 records), mortality rates (35% to 60%), and study designs. Our results indicate that for larger datasets (over 10,000 cases), including a synthetic patient population in the nearest network neighbor prediction model consistently improves performance compared to real patients alone. Specifically, AUROCs for cancers like cervix uteri and ovarian cancer ranged from 0.828 to 0.884 with synthetic patients, compared to 0.720 to 0.858 with real data. These findings highlight the advantages of network-based digital twins and the importance of effective sample size in predictive modeling.

Methods

Cancer Registry Data

Data retrieval is from SEER Stat Version 8.4.1, filtering for unique cases from 2010 to 2015 with vital status indicating alive or dead due to cancer. Our criteria included a sample size range of 1,000 to 30,000 records, ensuring a diverse representation of deceased cases (35%-60%). The study focused on seven cancers: Ovary (n=30,699), Cervix Uteri (n=17,855), Small Intestine (n=9,942), Hepatic Flexure (n=5,204), Nasopharynx (n=2,990), Oropharynx (n=2,050), and Other Urinary Organs (n=1,308). Features included age, year of diagnosis, sex, race, ICDO3, tumor grade, laterality, primary site, survival in months, tumor sequence, diagnostic confirmation, ICCC site, combined summary stage, and vital status (Alive or Dead). The last feature was used as the clinical outcome of class variable for prediction.

Study Design and Analysis.

The data processing steps are outlined in the flowchart in Figure 1.
We compare the outcome predictions of seven cancers to the baseline of breast cancer using the SynTwin algorithm through the following steps:

Synthetic Data Generator

Mixture of product of multinomials (MPoM) synthetic data generation algorithms are used to generate synthetic patients.

Network Analysis

This network is built with patients as nodes and edges with weights based on the estimated distances. An inflection point is calculated and applied to filter edge distances. We evaluated Gower distance metrics.

Community Distance Selection

SynTwin uses the Multilevel algorithm to identify large communities, optimizing resolution parameters to maximize those with at least 10 nodes. For each community, digital twins are selected based on the distance from the central node to the farthest node within that community.

Nearest Neighbor Prediction

Mortality (Alive or Dead) of the holdout patients is predicted using a majority vote. We tested the prediction performance using six study designs (Figure 2).

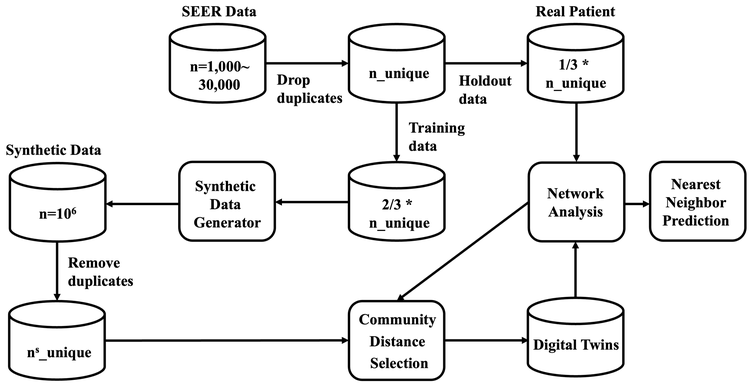

Figure 1_ Flowchart for data processing and analysis.

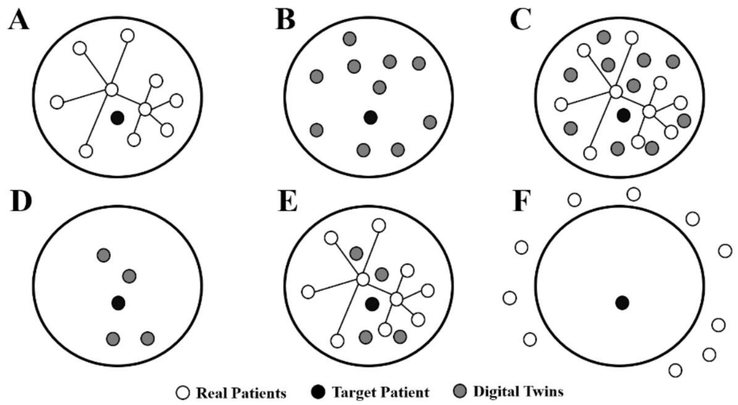

Figure 2_ Study design for comparing outcome prediction using real patients and/or digital twins (Moore et al., 2024). The large circles represent a community within the patient network.

Results

Table 1 presents an overview the AUROC performance of SynTwin for predicting mortality in the holdout data across various cancer sites for each study design (A-F). As sample size increased, the range of confidence intervals narrowed down, indicating improved precision in the estimates. Interestingly, SynTwin exhibited consistent performance across cancers with large sample sizes, irrespective of the variation in the percentage of mortality among different cancer types, from 35% for small intestine to 58% for ovarian cancer.

Table 1_ Comparison of SynTwin performance as measured by AUROC on various cancer datasets. Bolded metric values are significantly better than the others.

Approximate sample size	Cancer site	Design*	A	B	C	D	E	F
90,000	Breast (Moore et al., 2024)	mean	0.791	0.784	0.783	**0.864**	**0.852**	0.494
30,000	Ovary	mean	0.858	0.848	0.842	**0.882**	**0.884**	0.431
20,000	Cervix Uteri	mean	0.720	0.708	0.702	**0.835**	**0.828**	0.496
10,000	Small Intestine	mean	0.783	0.772	0.761	**0.872**	**0.861**	0.468
5,000	Hepatic Flexure	mean	0.768	0.789	0.852	0.780	0.844	0.554
3,000	Nasopharynx	mean	0.597	0.594	0.578	**0.708**	**0.709**	0.552
2,000	Oropharynx	mean	0.671	0.683	0.664	0.746	0.768	0.438
1,000	Other Urinary Organs	mean	0.467	0.530	0.466	0.670	0.635	0.522

Discussion

Despite the robust performance of SynTwin in predicting mortality across cancer types with sample sizes exceeding 10,000 cases, there are several considerations for future research remain.

- First, the scalability of this approach when applied to complex datasets containing a large number of features with a mix of numerical, categorical, and binary features remains uncertain, as this was not addressed in the current study. The synthetic dataset generator used in this study (MPoM) only models categorical features, and therefore cannot be directly applied to numerical features.
- Second, implementing SynTwin for large datasets with more features might be computationally intensive, so optimizing the method to maximize utility with limited computing resources is essential.

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